

PrimeTrade Data Science Internship Assignment

Analyzing the Relationship Between Bitcoin Market Sentiment and Hyperliquid Trader Performance

Submitted by

Ananya Verma

1. Objective

The objective of this project was to analyze whether Bitcoin market sentiment influences trader behavior and trading performance on the Hyperliquid platform.

By combining the Bitcoin Fear & Greed Index with historical trading records, the project investigates how sentiment relates to profitability, leverage usage, trade size, and overall trading activity.

2. Datasets Used

Bitcoin Fear & Greed Index

Contains daily market sentiment classified as Fear or Greed.

Hyperliquid Historical Trader Data

Contains historical trading records including:

- Execution Price
 - Trade Size
 - Trading Side
 - Closed PnL
 - Leverage
 - Account Information
 - Trading Timestamp
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3. Methodology

The following workflow was followed:

- Imported and explored both datasets
- Cleaned missing and inconsistent values
- Converted timestamps into comparable dates
- Merged both datasets using trading date
- Performed exploratory data analysis

- Created visualizations
 - Compared trading metrics across different sentiment conditions
 - Conducted statistical analysis to validate observed trends
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4. Analysis Performed

The project includes analysis of:

- Distribution of market sentiment
 - Average Closed PnL
 - Win Rate
 - Leverage Comparison
 - Trade Size Analysis
 - Long vs Short Position Distribution
 - Most Frequently Traded Assets
 - Correlation Analysis
 - Statistical Significance Testing
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5. Key Findings

The analysis indicates that market sentiment influences trading behaviour and risk-taking patterns.

Differences were observed in profitability, leverage usage, and trading activity across Fear and Greed market conditions. While sentiment alone does not determine trading success, it provides valuable context that can complement quantitative trading strategies and improve risk management.

6. Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Incorporate market sentiment indicators into trading strategies.
 - Monitor leverage carefully during periods of strong market optimism.
 - Combine sentiment data with technical indicators for better decision-making.
 - Apply stricter risk management during highly volatile market conditions.
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7. Limitations

This analysis has several limitations:

- Market sentiment is only one factor affecting trader performance.

- Results depend on the quality and completeness of the provided datasets.
 - External macroeconomic events and market news were not considered.
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8. Future Improvements

Future work could include:

- Machine Learning models for trader profitability prediction
 - Time-series forecasting
 - Portfolio-level performance analysis
 - Real-time sentiment integration
 - Advanced clustering of trader behaviour
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9. Conclusion

This project successfully explored the relationship between Bitcoin market sentiment and Hyperliquid trader performance using Python-based data analysis techniques.

The findings demonstrate that market sentiment offers useful insights into trader behaviour and can serve as an additional factor in developing data-driven trading strategies.

A few final suggestions before you